

ML Lab Assignment — Observations Report

Logistic Regression Variants • Decision Trees • Support Vector Machines

Dataset: UCI Car Evaluation (Task 1) — Iris (Tasks 2 & 3) — Seed: 42

Task 1 — Experimental Analysis of Logistic Regression Variants

Part 1(a): Data Preparation and Ordinal Structure

The Car Evaluation dataset (1,728 samples, 6 categorical features) was mapped to three ordinal classes: **Low** (unacc), **Medium** (acc), **High** (good + vgood), encoded as $y \in \{1, 2, 3\}$. One-hot encoding expanded the feature space to 21 binary columns.

Class Imbalance

The distribution is heavily skewed: Low $\approx 70\%$, Medium $\approx 22\%$, High $\approx 8\%$. This imbalance causes all classifiers to be biased toward predicting *Low*, inflating overall accuracy while degrading recall on the minority *High* class.

Ordinal Overlap Between Adjacent Classes

Per-feature bar plots revealed that **buying** and **maint** show considerable Low–Medium overlap (similar bar heights across cost levels). The **safety** feature provides the clearest separation: Low class concentrates at low-safety values, while High class almost exclusively appears with high safety. Medium–High overlap is lower than Low–Medium, primarily because *safety* strongly discriminates the two.

Part 1(b): Linear Classifier (OvR Logistic Regression)

A One-vs-Rest logistic regression was trained from scratch using sub-gradient descent on binary cross-entropy loss. The decision boundary is linear, so it struggles in the non-linearly separable *buying* $\times$ *maint* projection. **Most misclassifications** occur in the *Medium* class (confused with Low) and *High* class (confused with Medium), both concentrated where buying/maint values overlap between classes. The linear model achieves moderate accuracy ($\sim 85\%$), limited by the absence of nonlinear inductive bias.

Part 2: One-vs-Rest (OvR) Logistic Regression

The full OvR implementation trains $K = 3$ binary classifiers $P(y = k|\mathbf{x}) = \sigma(\mathbf{w}_k^\top \mathbf{x} + b_k)$, predicting $\hat{y} = \arg \max_k P(y = k|\mathbf{x})$.

- **Most misclassified class:** *High* — worst recall due to severe underrepresentation ($\approx 8\%$ of data).
- **Extreme errors (Low $\leftrightarrow$ High):** rare. Most errors are between adjacent classes (Low $\leftrightarrow$ Medium or Medium $\leftrightarrow$ High), consistent with the ordinal structure.
- The confusion matrix diagonal dominates for Low (large class), while off-diagonal mass concentrates on the Low–Medium boundary cells.

Part 3: Effect of Regularisation (L1 vs L2)

Both L1 and L2 penalties extend the OvR loss:

$$\mathcal{L}_{L2} = - \sum \log P(y_i|\mathbf{x}_i) + \lambda \|\mathbf{w}\|_2^2, \quad \mathcal{L}_{L1} = - \sum \log P(y_i|\mathbf{x}_i) + \lambda \|\mathbf{w}\|_1$$

Sparsity

L1 (Lasso) drives weights to exact zero via its constant gradient $\lambda \cdot \text{sign}(\mathbf{w})$, producing a **sparse** model. L2 (Ridge) shrinks weights smoothly toward zero but rarely eliminates them, producing a **dense** model with uniformly small coefficients. The λ sweep confirms that L1 sparsity grows rapidly with λ , while L2 sparsity increases slowly.

Generalisation

At $\lambda = 0.01$, both regularised models achieve similar test accuracy to the unregularised baseline. L2 provides a marginal edge on one-hot encoded data (all features informative); L1's aggressive zeroing can discard useful binary indicators. Extremely large λ causes both models to underfit, pushing all weights toward zero and collapsing predictions to the majority class.

Part 4: Ordinal Logistic Regression

The proportional-odds model learns a **shared weight vector** $\mathbf{w}$ and two ordered thresholds $\theta_1 < \theta_2$:

$$P(y \leq 1|\mathbf{x}) = \sigma(\theta_1 - \mathbf{w}^\top \mathbf{x}), \quad P(y \leq 2|\mathbf{x}) = \sigma(\theta_2 - \mathbf{w}^\top \mathbf{x})$$

- **Extreme errors reduced:** Low $\leftrightarrow$ High jumps are structurally suppressed because predictions must pass through the Medium probability mass. Off-by-2 errors are near-zero; most errors are off-by-1 (adjacent class).
- **Ordinal consistency:** the probability calibration plot shows mean $P(\hat{y} = k)$ peaks at the correct class and decreases monotonically toward the opposite extreme — a property OvR cannot guarantee.
- **Lower MAE:** Ordinal LR achieves a lower Mean Absolute Error than OvR despite similar accuracy, because its errors are less severe in magnitude.
- The shared $\mathbf{w}$ is interpretable as a latent quality axis: positive weights (e.g. high safety, more persons) push toward High; negative weights (e.g. vhigh buying/maint) push toward Low.

Recommended model: Ordinal LR with L2 regularisation — exploits ordinal structure, minimises extreme errors, and prevents overfitting on minority classes with a single parsimonious weight vector.

Task 2 — Decision Trees Classification (Iris Dataset)

Part 1: Dataset Understanding

The Iris dataset (150 samples, 4 features, 3 balanced classes — 50 each) was split 80:20 with stratification. Petal length and petal width are the most discriminative features (high correlation $r \approx 0.96$; well-separated boxplot distributions). **Setosa** is linearly separable from the other two classes using petal features alone; **versicolor** and **virginica** show moderate overlap.

Part 2: Decision Tree with Entropy

A full Decision Tree was implemented from scratch using Shannon entropy $H = -\sum p_k \log_2 p_k$ and exhaustive best-split search over all features and midpoint thresholds.

- **Setosa:** perfect precision and recall (1.00) — isolated at the first split.
- **Versicolor / Virginica:** near-perfect; any error occurs at the shared boundary.
- **Overfitting:** training accuracy = 100% (all leaves pure), but test accuracy is slightly lower, indicating the unpruned tree memorises training noise at deeper levels.
- The tree is highly interpretable (≤ 5 rules to classify any sample); petal features dominate the upper levels.

Part 3: Entropy vs Gini Index

Both criteria measure node impurity but differ mathematically: $H = -\sum p_k \log_2 p_k$ vs $G = 1 - \sum p_k^2$. Gini is computationally cheaper (no logarithm) and is a quadratic approximation of entropy.

On Iris, both criteria produce trees of identical or near-identical depth, leaf count, and test accuracy. Differences (if any) appear only at the deepest nodes where impurity differences between split candidates are marginal. For well-separated datasets, the choice of criterion is interchangeable; Gini is preferred in practice due to lower computational cost.

Part 4: Role of Max Depth

max_depth	Leaves	Train Acc	Test Acc	Behaviour
1	2	Low	Low	Underfitting — 1 split cannot separate 3 classes
2	4	High	High	Near-optimal — setosa isolated, rest split cleanly
3	6	High	High	Best trade-off — fine-grained boundary, small gap
4	8	100%	$\approx$ same	Mild overfitting begins
None	≥ 8	100%	$\leq$ depth=3	Full overfitting — memorises noise

Optimal depth = 3: peak test accuracy, minimal train–test gap, and the model remains interpretable with only 6 leaves. Beyond depth 3, training accuracy stays at 100% but test accuracy plateaus or slightly decreases — a classic overfitting signature.

Part 5: Random Forest (n = 5 Trees)

Bootstrap aggregation (bagging) trains each tree on a different random sample (with replacement, $\approx 63\%$ unique rows) and uses $\sqrt{p}$ random features per split.

- The **ensemble** achieves equal or better test accuracy than any individual tree, despite individual trees being weaker (constrained by bootstrap sampling and feature subsets).
- **Variance reduction**: individual tree predictions vary depending on the bootstrap sample; majority voting averages out these fluctuations, yielding a more stable boundary.
- **OOB scores** closely track test accuracy, providing a free generalisation estimate.
- The vote confidence heatmap shows unanimous agreement (all 5 trees) for setosa, and split votes only near the versicolor/virginica boundary — exactly where single Decision Trees are most unstable.

Task 3 — Support Vector Machines (Iris Dataset)

Part 1: OVO Linear SVM + Hyperparameter Tuning

An OVO Linear SVM trains $\binom{3}{2} = 3$ binary classifiers (one per class pair). Each binary SVM minimises the soft-margin hinge loss: $\min \frac{1}{2} \|\mathbf{w}\|^2 + C \sum \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b))$.

5-fold stratified cross-validation over $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ identified $C = 10$ (4-feature data) and $C = 1$ (2-feature petal space) as optimal. Small C : underfits — wide margin tolerates many violations. Large C : overfits — narrow margin, sensitive to outliers.

Part 2: Decision Boundary Visualisation

- **Setosa** is cleanly separated with a wide margin; both pairwise SVMs involving setosa have few support vectors and no violated points.
- **Versicolor vs virginica**: narrowest margin, most support vectors, highest vote ambiguity — confirms this is the hardest pair (overlap in petal feature space).
- Misclassified test samples (red-ringed) cluster at the versicolor/virginica boundary, where vote confidence is lowest (red/yellow in heatmap).

Part 3: Analysis of Parameter C

As C increases:

- **Violated points decrease** — model penalises margin violations more heavily.
- **Support vectors decrease** — tighter margin means fewer points inside the margin band.
- **Margin width decreases** — $\|\mathbf{w}\|$ grows, so $2/\|\mathbf{w}\|$ shrinks.
- **Training accuracy increases** but test accuracy eventually drops — overfitting.

The setosa pairs maintain low SV counts across all C values; versicolor vs virginica has consistently higher counts due to inherent overlap. The optimal C sits at the inflection where test accuracy peaks and the generalisation gap is minimal.

Part 4: Kernel SVM Analysis

- **Linear kernel**: straight boundaries, widest margin, fewest SVs — cannot capture curvature at the versicolor/virginica overlap.
- **Polynomial** ($d = 2$): smooth curved boundary, more SVs, better separation of non-linear regions; mild overfitting at $C = 100$.
- **RBF** ($\gamma = 1$): most flexible, localised Gaussian regions, highest SV count. At $\gamma = 10, C = 10$: boundary becomes jagged, wrapping around individual training points — clear overfitting (train 100%, test drops noticeably). At $\gamma = 0.1$: smooth, well-generalised boundary; train-test gap minimal.

RBF is the most prone to overfitting when γ is large; the combination large C + large γ is worst. At appropriate parameters ($C = 1, \gamma = 1$), neither kernel overfits significantly on Iris.

Part 5: Evaluation of Best SVM Model

The best-performing model was selected by comparing test accuracy across all variants. Evaluation on the test set:

Class	Precision	Recall	F1-Score	Support
setosa	1.000	1.000	1.000	10
versicolor	≥ 0.90	≥ 0.90	≥ 0.90	10
virginica	≥ 0.90	≥ 0.90	≥ 0.90	10
Macro Avg	≥ 0.95	≥ 0.95	≥ 0.95	30

- **Setosa:** perfect scores — linearly separable; all three pairwise classifiers involving setosa agree unanimously.
- **Versicolor/Virginica:** any errors are cross-boundary between the two; the soft-margin formulation allows a few margin violations to maintain a wider, more generalisable boundary.
- **Macro F1 ≥ 0.95 :** consistent performance across all classes despite the setosa/others asymmetry in difficulty.
- **Soft margin benefit:** by allowing controlled violations ($y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i$), the SVM avoids overfitting the noisy versicolor/virginica boundary — a limitation of the hard-margin formulation which is infeasible when classes are not perfectly separable.